VINCENZO CECCARELLI GRIMALDI

PhD Applicant — ERC Synergy Grant SAFEr Grid (JobID: V000010837)

CONTACT

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CORE STRENGTHS

Energy Systems Modeling
Physics-informed Digital Twins, Cyber-Physical Energy Systems Modeling & Simulation, Smart Grid Stability Analysis, Asynchronous Subgrid Operation, DER Integration, Predictive Modeling, Real-Time Co-Simulation

Intelligent Control
Reinforcement Learning for Adaptive Grid Control, Multi-Agent Systems, Model Predictive Control (MPC), Agent-Based Stability Optimization, Python (PyTorch, Pandas, NumPy, Scikit-learn), Matlab/Simulink, Git, Linux, Docker

INTERESTS

- Cyber-Physical Power Systems
- Physics-Informed Digital Twins
- Smart Grid Stability & Dynamics
- Asynchronous Operation of Modular Subgrids
- Agent-Based Control & Coordination
- Decentralized Energy Resource Management
- Real-Time Simulation & Adaptive Control
- Store-and-Forward Energy Architectures

ACCREDITATION

ITK Fachspezialist
DB InfraGO AG
01 Jan 2026 — Berlin

LANGUAGES

- German — Professional
- English — Professional
- Spanish — Professional
- Chinese Mandarin — Professional

PROFESSIONAL SUMMARY

As an Electrical Business Engineer I have developed a strong background in physics-informed digital twins, reinforcement learning for adaptive control, and AI-driven energy systems modeling. I develop platform architectures that enable resilient, interoperable, and decentralized smart grids – with a particular focus on asynchronous operation of modular subgrids and energy flows. My interdisciplinary business-engineering perspective enables the seamless translation of technical solutions into advanced cyber-physical control strategies and stability-oriented architectures for future power systems.

SELECTED INNOVATION & PLATFORM PROJECTS

- **GridOS**: Physics-informed digital twin platform with multi-protocol adapters and agent-based simulations for stability optimization and decentralized energy flows in asynchronously coupled smart-grid architectures
- **NeuralBridge**: Adapter platform for autonomous AI agents and cross-domain data exchange – enabling coordinated energy and information flows in modular subgrid systems
- **Reinforcement Learning Integration** (GridOS ↔ NeuralBridge): Integration of reinforcement-learning models into both platform architectures for real-time simulation and stability analysis of decentralized energy systems

EDUCATION

M.Sc. Business Administration & Engineering: Electrical Power Engineering
Overall Grade: 2.7 – Master's thesis 2.3
Aug 2022 – Aug 2025 — RWTH Aachen University, NRW

- **Master's thesis (Grade 2.3):** Data Modeling in a Cross-domain Ontology for Cyber Intelligence in Smart Grids Using Reinforcement Learning

B.Sc. Business Administration & Engineering: Electrical Power Engineering
Overall Grade: 3.5 – Bachelor's thesis 3.0
Sept 2018 – Aug 2022 — RWTH Aachen University, NRW

- **Bachelor's thesis (Grade 3.0):** Development of a Cross-domain Knowledge Base for Cyber Intelligence in Smart Grids

RELEVANT WORK EXPERIENCE

ITK Specialist
DB InfraGO AG - Technische Experten / Wissensmanagement 1 (V.IO-MI-IW 1)
Aug 2024 – Present — Frankfurt am Main, HE

- Designed and implemented reinforcement-learning agents for adaptive control and real-time stability optimization in electrified subway power infrastructures under variable load conditions using Python and PyTorch
- Developed and integrated multi-protocol, agent-based digital control systems for critical infrastructure applications, including passenger information, emergency communication and voice alarm systems
- Conducted technical plan reviews, system commissioning and compliance validation according to DIN VDE 0833-4 and DB standards to ensure reliable operation of high- and medium-voltage systems

Industrial Internships (Betriebspraktikum)
DB Fahrzeuginstandhaltung GmbH - Vertriebliches Angebotsmanagement und Vertragsadministration (P.WVV 34)
Dec 2022 – May 2024 — Frankfurt am Main, HE

- Performed dielectric testing and ultrasonic flaw detection on high-voltage roof insulators to evaluate insulation integrity and flashover risks in operational high-voltage components
- Optimized electrical stability and predictive maintenance strategies for pantograph integration systems through precision alignment of roof bushings and pneumatic feeds

Industrial Internship

DB Netz AG - Supplier and System Testing IH (V.IWM 17)

Aug 2022 – Dec 2022 | Frankfurt am Main, GER

- Supported quality assurance and functional testing of GSM-R communication components for safety-critical railway signaling systems
- Performed time-series analysis of train run data and schedule deviations using the LeiDis-NK dispatching system to enhance operational stability and system reliability